



Course:	Data Structures and Algorithm Lab	Course Code:	CL214
Program:	BS(EE)	Semester:	Spring 2017
Duration:	120 Minutes	Total Marks:	30 Absolute
Paper Date:	3-May-17		
Section:	EL 2A and EL 2B	Page(s):	2
Exam:	Lab Final		

Roll Number:

Instruction/Notes:

Merge sort is an efficient, general-purpose, comparison-based sorting algorithm. Mergesort is a divide and conquer algorithm that was invented by John von Neumann in 1945. Conceptually, a merge sort works as follows:

Divide the unsorted list into n sub lists, each containing 1 element (a list of 1 element is considered sorted).

Repeatedly merge sub lists to produce new sorted sub lists until there is only 1 sub list remaining. This will be the sorted list.

Following figure represents different steps of the merge sort algorithm.

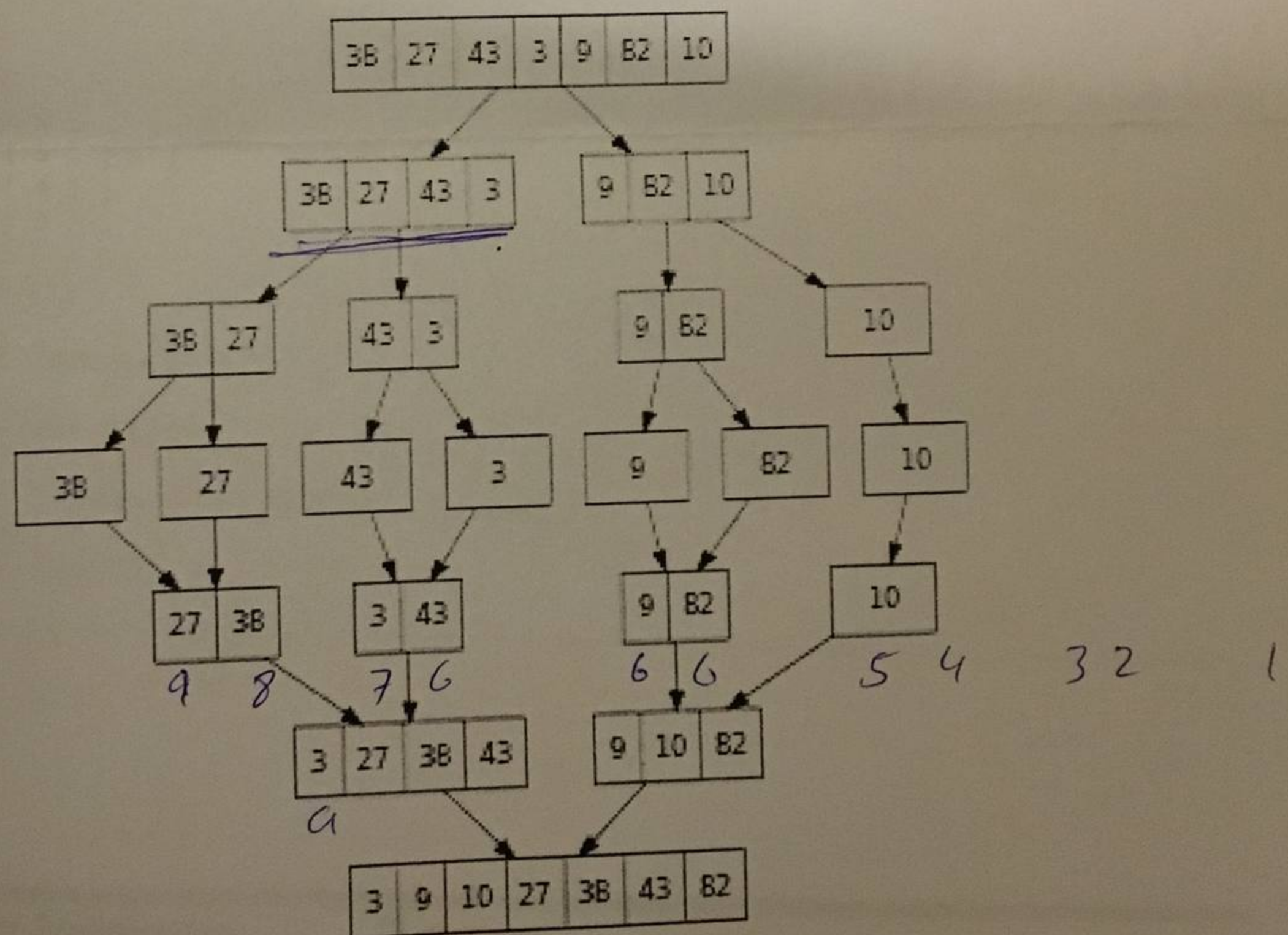


Figure 1: Merge Sorting Algorithm example

National University of Computer and Emerging Sciences, Lahore Campus



Course:	Data Structures and Algorithms	Course Code:	CS210
Program:	BS (Electrical Engineering)	Semester:	Spring 2017
Assessment Tool	Quiz # 3	Date:	03-05-17
Total Marks:		Roll # :	
Name :			

Instructions: This is a closed-book, closed-notes, closed-computer, no-neighbor-peek quiz.

Question 1:

[CLO 04]

Consider the problem of sorting, in worst-case linear time, an array A of 10,000 9-digit NADRA numbers in increasing order. For each of the sorting algorithms below, indicate whether or not the algorithm will achieve worst-case, linear-time performance, and briefly explain why or why not.

[6 Marks]

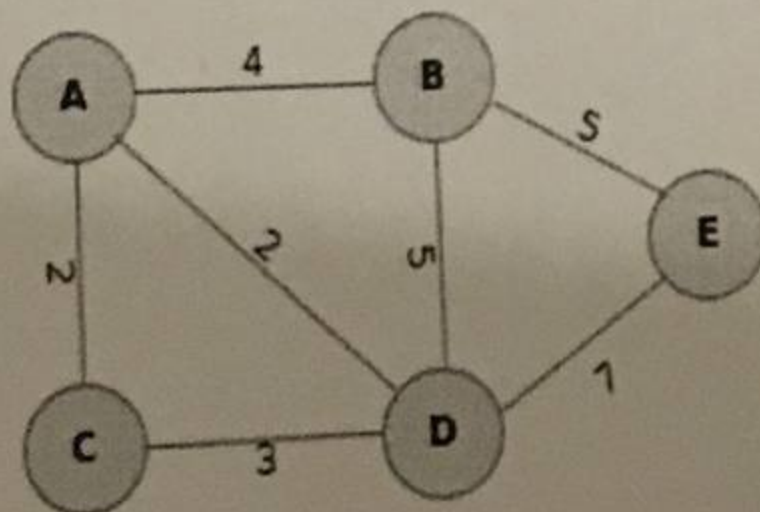
a) Quick Sort

b) Bucket/Bin Sort

c) Radix Sort

Question 2:

[CLO 04]
[10 Marks]



- i) Consider the graph given above. Which of the following statements is TRUE?
- A. Minimum spanning tree includes edge $\langle C, D \rangle$.
 - B. Prim's algorithm is not application in the given graph to calculate minimum

spanning tree.

C. The given graph is a directed graph of total weight 28.

D. none of the above

ii) Consider the graph given above. What is the weight of Minimum Spanning Tree of the given Graph.

A. 13

B. 12

C. 14

D. 16

iii) What graph traversal algorithm uses a queue to keep track of vertices which need to be processed? Mark the correct statement.

A. Breadth First Search

B. neither BFS nor DFS

C. Depth First Search

D. both DFS and BFS

iv) If G is a directed graph with 20 vertices and all edges having unit weight, how many boolean values will be needed to represent G using an adjacency matrix? Explain briefly.